

TECHNICAL REPORT

LEXO – 4IELTS

A bilingual (English + Arabic) IELTS preparation platform — flashcards, mock tests, AI-graded writing, AI-assisted speaking, listening practice, and structured study plans.

Production: lexo.4ielts.com **Generated:** 2026-04-29

Stack: pnpm monorepo · TypeScript · React 19 · Express 5 · PostgreSQL

1. Project Overview

LEXO is a comprehensive IELTS prep web app aimed at Arabic-speaking learners worldwide. It blends a vocabulary-first study system with full exam practice (Listening, Reading, Writing, Speaking), AI tutoring, structured daily plans, and a teacher dashboard for monitoring students. The platform is delivered as a Progressive Web App so students can install it to their home screen.

2. Repository Architecture

The codebase is a **pnpm monorepo** with three deployable artifacts and several shared libraries.

Artifacts

- **flashcards** Student-facing React 19 + Vite SPA (the LEXO web app).
- **api-server** Express 5 backend, all REST endpoints under `/api`.
- **mockup-sandbox** Internal design preview server for component prototyping.

Shared libraries

- **@workspace/db** Drizzle schema + migrations.
- **@workspace/api-spec** OpenAPI source of truth.
- **@workspace/api-zod** Generated Zod validators.
- **@workspace/api-client-react** Generated React Query hooks.
- **@workspace/integrations-anthropic-ai** Claude integration helper.

Routing across services is handled by a global path-based reverse proxy: requests under `/api/*` reach the API server; everything else reaches the web app. The OpenAPI contract is the source of truth — running codegen produces the Zod validators on the server and the React Query hooks on the client, keeping the contract end-to-end type-safe.

3. Backend (api-server)

Layer	Technology
Runtime	Node.js 24, ES modules, esbuild bundle
Framework	Express 5
Database	PostgreSQL via Drizzle ORM
Validation	Zod (auto-generated from OpenAPI)
Logging	Pino + pino-http (structured request logs)
Auth	Email + access code · bcrypt hashing · client-stored session token · idle timeout
AI	Anthropic Claude (writing & chat) · OpenAI TTS · Orwell AI (essay grading)
PDFs	puppeteer-core + system Chromium (cached on disk)
Notifications	VAPID web-push, in-app + browser push
Scheduling	node-cron (daily reminders, AI-usage rollups)
File uploads	multer + heic-convert (avatars, attachments)

Route groups

- auth
- flashcards
- lessons
- essay-check
- orwell
- lexo-ai
- speaking
- stories
- story-exercises
- notifications
- plan-pdf
- vocab-pdf
- sentence-check
- sentence-sessions
- spell-it
- admin-ai-usage
- admin-data-fixes
- admin-notifications
- health

4. Database Schema (highlights)

Table	Purpose
flashcards	2,198 CEFR-graded vocabulary cards with Arabic translations and bilingual examples.
progress · bookmarks	Per-user learning state and saved cards.
card_srs	Spaced repetition (SM-2 algorithm) — interval, ease, due date.
activity_positions	Resume-where-you-left-off across all study modes.
quiz_scores	Quiz history and analytics.
weak_words	Auto-curated review deck of words the user struggles with.
xp_events	Gamification — every XP-earning action is recorded.
user_data	Profile, plan duration, settings, avatar.
push_subscriptions	VAPID web-push endpoints per device.
ai_usage_events · admin_alerts_sent	Per-call AI logging + cost-spike admin email alerts (SendGrid).
reviews	Public student testimonials with admin reply support.

5. Frontend (flashcards artifact)

Stack • React 19, Vite 7, TypeScript strict • TanStack Query (data layer) • Tailwind v4 + Radix UI primitives • Framer Motion (animations) • Wouter (routing) · next-themes (dark mode) • react-hook-form + Zod (forms) • Embla carousel · cmdk · input-otp · date-fns • PWA installable, browser TTS, RTL Arabic support	Engagement layer • XP system + daily streaks • Weak-word recycling deck • Guided onboarding tour • Idle warning + auto-logout modal • Floating LEXO AI chat bubble • Public reviews on landing page • Web push notifications
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Major pages

- Home
- Onboarding
- Daily Plan
- Study
- Quiz
- Browse
- Spell It
- Flip It
- Sentence Builder
- Reading Test
- Listening Test
- Mock Test
- Speaking
- Speaking Topics
- Essay Checker

6. Reading Skills Practice (current focus)

Targeted exercises for each of the **17 IELTS Reading question types**, organised across **3 CEFR levels (A2 / B1 / B2)**. The newest type is **Coherence and Cohesion**:

- 3 levels × 20 questions each (10 coherence + 10 cohesion).
- Per-question A/B/C/D options rendered as full-text choice cards.
- Brief italic “Why:” explanation under every item on the results screen.
- Partial submission allowed — students can submit at any time.
- Skipped items shown in amber, correct in green, incorrect in red.
- XP reward scales for the longer practice: 2 XP per correct, capped at 40.

7. AI Capabilities

Feature	Model / Service	Use
Essay grading (Orwell AI)	Anthropic Claude	Band scoring + feedback for IELTS writing tasks.
LEXO AI Chat	Anthropic Claude	In-app IELTS tutor (vocab, grammar, exam tips).
AI story exercises	Anthropic Claude	Generated comprehension questions for stories.
Pronunciation	OpenAI TTS (onyx, tts-1-hd)	Spoken word + example sentences.
Quiz generation	Faster Claude tier	On-demand question generation.

Every successful AI call is recorded in `ai_usage_events` . A nightly job aggregates spend; if usage crosses thresholds, an alert email is dispatched via SendGrid.

8. Quality, Safety & Operations

- **Type safety end-to-end** — root `pnpm run typecheck` builds composite libs first, then leaf apps.
- **Contract-first APIs** — OpenAPI generates Zod (server) and React Query hooks (client).
- **Server logging** — `req.log` in handlers, singleton `logger` elsewhere; `console.log` is forbidden in server code.

- **Admin-only writes** — destructive prod fixes go through password-gated admin endpoints (header check).
- **Idle session timeout** with warning modal; tokens validated server-side on bootstrap.
- **Build & deploy** — Replit hosts each artifact as its own workflow; the global proxy routes by path; HTTPS is automatic.

9. Recent Highlights (last ~50 commits)

- Rebranded landing & registration headline to **“LEXO - 4IELTS”**.
- Added **Coherence and Cohesion** reading practice; total reading question types now **17**.
- Reordered Reading Test to put Skills Practice first, Full Tests second.
- Decoded HTML entities in 41 production flashcard sentences via the admin endpoint.
- Added Skimming and Scanning question types with tiered difficulty levels.
- Improved answer matching to accept common spelling variants.
- Added AI-powered story exercises and improved streak tracking.
- Faster quiz generation by switching to a lighter AI tier.
- Added video embedding to registration / login pages.
- Added “mark as weak” across all study modes; fixed study-session resume blank-screen bug.
- Profile page with avatar upload, name editing, and password change.
- Server-rendered bilingual EN+AR study-plan PDF download.
- Manual audio playback buttons on flashcards.

10. Hosting

- Production domain: **lexo.4ielts.com**.
- Each artifact runs as an independent workflow under a global reverse proxy.
- Automatic HTTPS, environment-managed secrets, scheduled cron jobs, and on-demand PDF rendering.